

Molhm Alasri

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EDUCATION

New Jersey Institute of Technology B.S. Electrical Engineering (Transferred). Relevant Coursework: Digital Design, Circuits & Systems I, Differential Equations	Newark, NJ Expected May 2029
Stevens Institute of Technology B.E. Computer Engineering. Dean's List Presidential Achievement Scholarship. GPA: 3.76. Relevant Coursework: Engineering Design, Programming & Algorithmic Thinking (C++), Sustainable Systems with Sensors	Hoboken, NJ Sep 2025 – May 2026

EXPERIENCE

PC Builder and Repair Technician Freelance - Built 30+ custom PCs and completed 100+ hardware repairs at 5–10 jobs per month at peak. - Diagnosed desktop and laptop issues; replaced RAM, SSDs, GPUs, and cooling components. - Tested each system for stability before returning it to the client.	Jul 2021 – Present Clifton, NJ
Sales Associate / Shift Lead The UPS Store - Managed opening and closing procedures, register reconciliation, and daily store operations. - Reorganized peak-hour task assignments to improve workflow and reduce customer wait times. - Assisted 20–30 customers per shift and processed insured shipments up to \$10,000.	Jun 2025 – Mar 2026 Clifton, NJ
Family Caregiver Provided 10–15 hours per week of mobility, transportation, and medication support for two siblings with disabilities while enrolled full time.	2022 – Present Clifton, NJ

PROJECTS

Software-Defined Radio ADS-B Receiver Python, RTL-SDR - Captured raw IQ at 1090 MHz (2 MSPS) and implemented preamble correlation and pulse-position demodulation to recover 112-bit Mode S extended squitter frames. - Validated frames with CRC-24 error detection and parsed ICAO addresses, recovering 64% of the addresses identified by dump1090 on identical captures. - Traced the 36% gap to fixed-threshold false detections near the noise floor; characterized the sensitivity/false-positive curve and selected an operating point.
Air Quality Monitor —VOC Detection - Sampled MQ-135 sensor data and displayed filtered air-quality readings on an OLED. - Implemented smoothing logic and threshold-based LED and buzzer alerts for elevated readings. - Designed and 3D-printed a ventilated enclosure for the sensor, display, and electronics.
Automated Plant-Watering System - Used soil-moisture readings to trigger water-pump cycles at 10-minute sampling intervals. - Programmed moisture thresholds and timing controls to prevent unnecessary watering. - Designed and 3D-printed a 5 x 5 x 10-inch enclosure for the sensors, pump, and wiring.

SKILLS

Languages: C++, Python, MATLAB
Embedded Systems: Arduino, analog sensor interfacing, OLED integration, real-time data acquisition
Electronics: Circuit prototyping, breadboarding, analog measurement, digital filtering
Hardware: PC assembly, system-level troubleshooting, diagnostics, component replacement
Design & Tools: SolidWorks, 3D printing, enclosure design
Additional: Arabic (fluent), communication skills, reliability testing, Microsoft Excel, written and verbal communication skills, decision-making, ai, equipment calibration, Microsoft Word

ACTIVITIES & LEADERSHIP

IEEE - Member
Commuter Student Organization - Member
Muslim Student Association - Member